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UCD DUBLIN
UCD Centre for Space Research
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C-Space
Presented by
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Optimising Biomass Mission Data for Oceanic & Cryospheric Remote Sensing

Introduction & Objective

Biomass, launched in April 2025, is the **first satellite with a P-band SAR radar**. It provides unique long-wavelength observations capable of penetrating beyond the Earth's surface[1]. Conducted at ESTEC, this research evaluates how this data can be applied to oceanographic and cryospheric Earth Observation, demonstrating opportunities to **maximise the scientific return** of ESA's newest Earth Explorer mission. Though its primary objective is to quantify forest carbon stocks, **this internship investigates whether 435MHz SAR can characterise ocean properties such as sea surface salinity (SSS) & temperature (SST) through physical manipulation of dielectric properties of seawater.**

Figs 1a & b. (Left) Biomass payload in Vega C Composite. (Right) Launch from Kourou Spaceport, French Guiana.

Scientific Theory

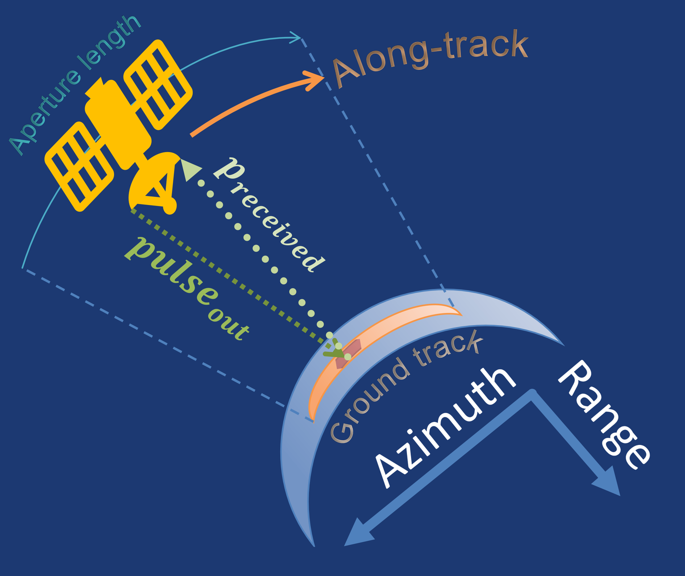


Fig 2. SAR Geometry: Pulses transmitted & received, defining azimuth & range.

- SAR systems transmit their own microwave pulses and measure the **phase** and **amplitude** of the return signal.
- **Comparing the *H*- and *V*-polarised returns directly gives the co-polarisation phase difference** (CPD), the key parameter observed throughout this work.
- **SSS and SST change seawater's dielectric constant, which changes how *HH* and *VV* reflect** pulses, so φ carries a SSS & SST signal, valid once surface roughness is corrected for. Higher φ correlates with increased SSS&T.[3]

$$P_r = \frac{P_t G^2 \lambda^2 \sigma^0 A}{(4\pi)^3 R^4} \cdot L$$

$$\varphi = arg(S_{HH} \cdot S_{VV}^*)$$

$$\frac{R_{VV}}{R_{HH}} \text{ depends on } \varepsilon(S,T)$$

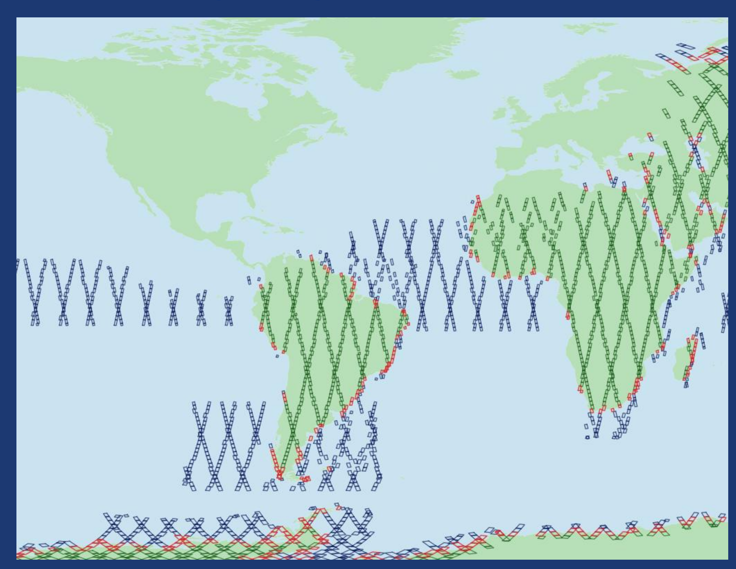
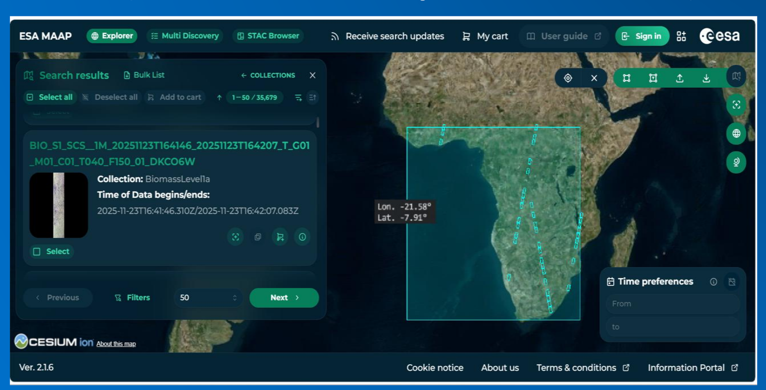


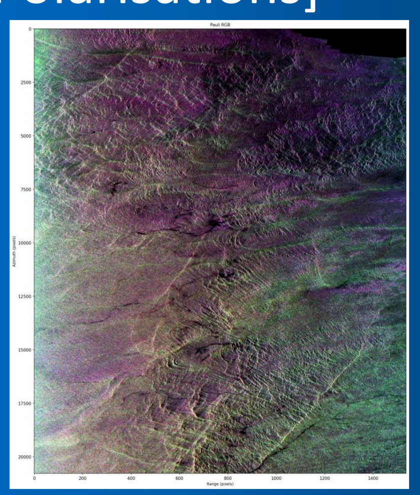
Fig 3. Limited data availability as only red (coastal) & blue (ocean) used

Methodology & Analysis

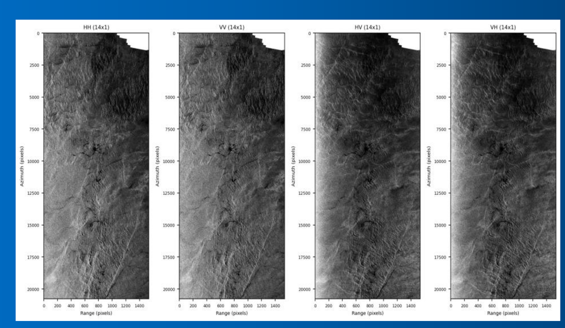
Download Biomass L1a Raw Products (Phase & Amplitude files)



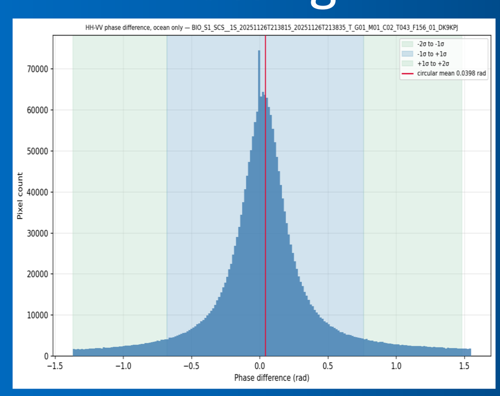
Check Baseline Visuals [Pauli RGB & Polarisations][2]



Land Masking & Multilooking to reduce speckle



CPD Min/Max Distribution & Full Swath Mosaic Plotting



Validation of SSS & SST [SMOS or SMAP] & [Sentinel3/Glorys]

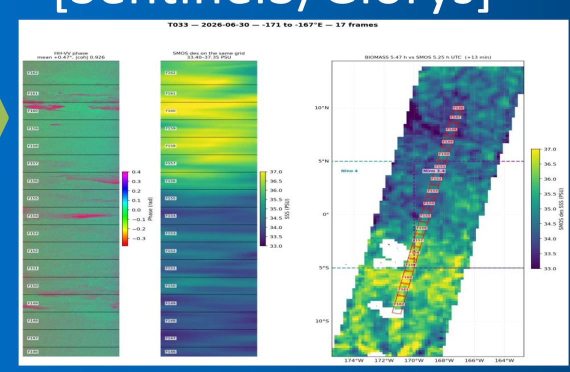


Fig 4. ESA MAAP Explorer, home of Biomass Data files for download

Fig 5a & 5b. Use of Python Notebooks to create Pauli Interpreted Coloured Plot & Grey Polarisation Plots with ESA WorldCover land mask (top right corner) example images used for product check

Fig 6. Histogram of CPD distribution used for min/max determination of 'useful' data

Fig 7. All products in swath in mosaic and validated against scene SMOS data

Results & Conclusions

- Results show Biomass's P-band polarimetry has clear resolution in CPD ocean region proving capabilities beyond forestry.
- **Salinity retrieval, though not yet a numerically quantified value, did show promising results in visual path strip data.**
- **Plume region showed** overall positive **correlation in CPD with SSS.** *(See Figures 8 & 9)*
- **El-Niño** case study found **visual correlation in both SSS & SST.** *(See Figures 10 & 11)*
- Together, these results open Biomass ocean applications as a promising new research direction.
- All results and code to support this study can be found by scanning below to access the dedicated Internship GitHub. *(See figure 12)*

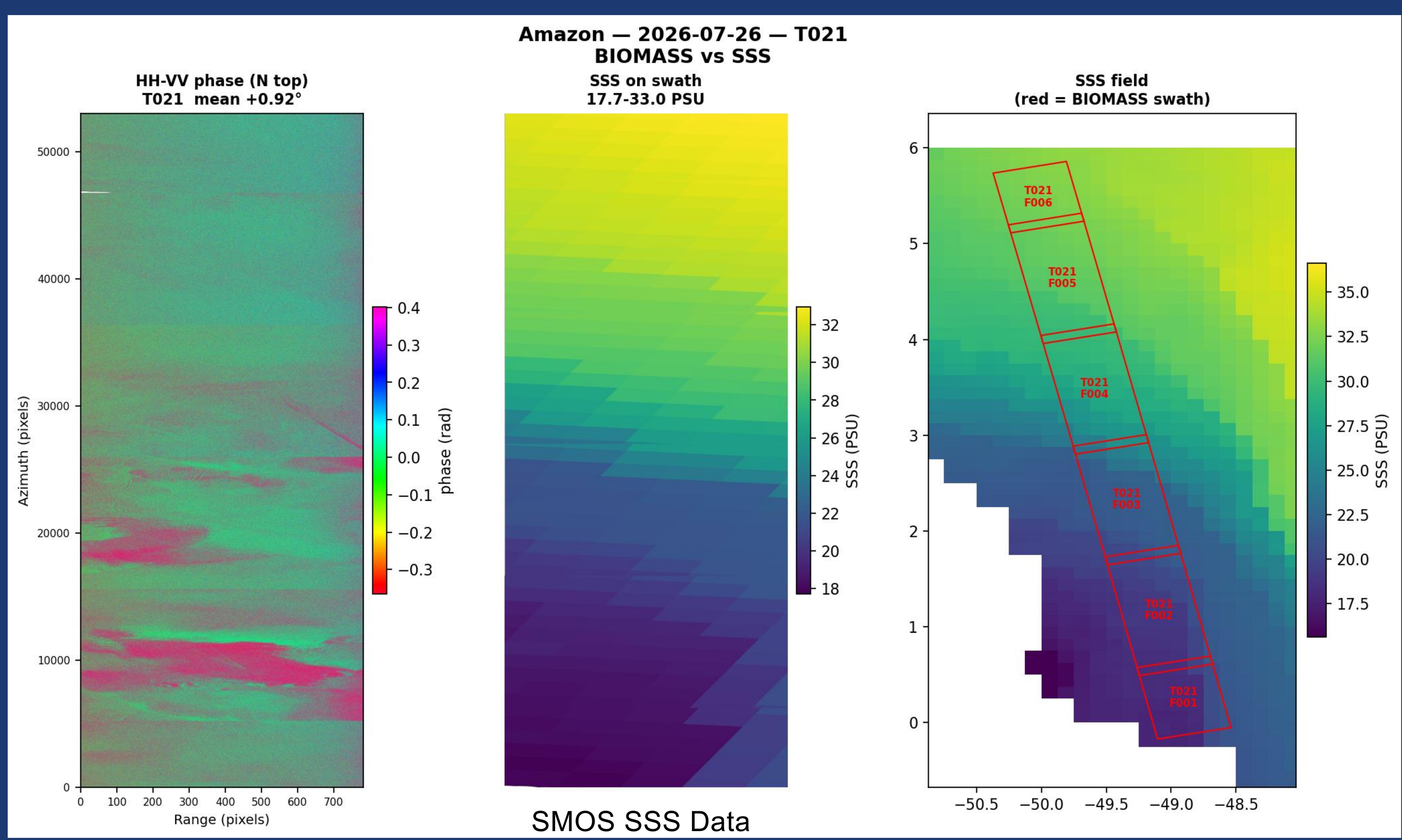


Fig 8. Visual results of CPD validated against SMOS Data for Amazon River Plume (Acquisition 26/07/26)

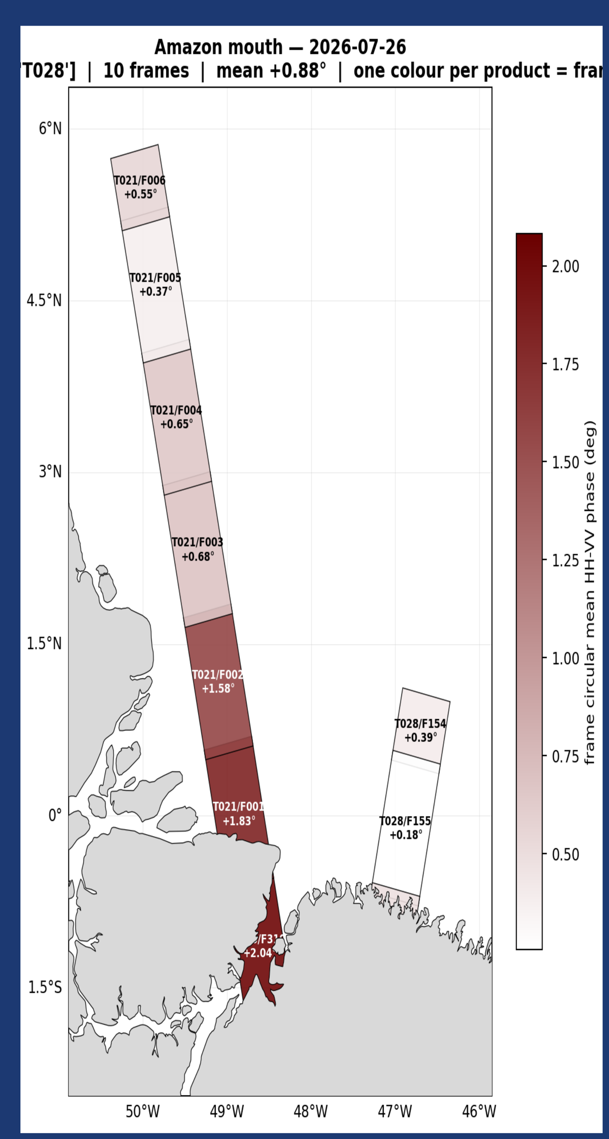


Fig 9. Product mean CPD values represented numerically with swath geometry

Fig 12. QR code for Git Repo

Future Work

Ocean Wave Dynamics
- Surface roughness
- Swell drift

Ice Applications
- Ground line detection
- Subglacial lakes

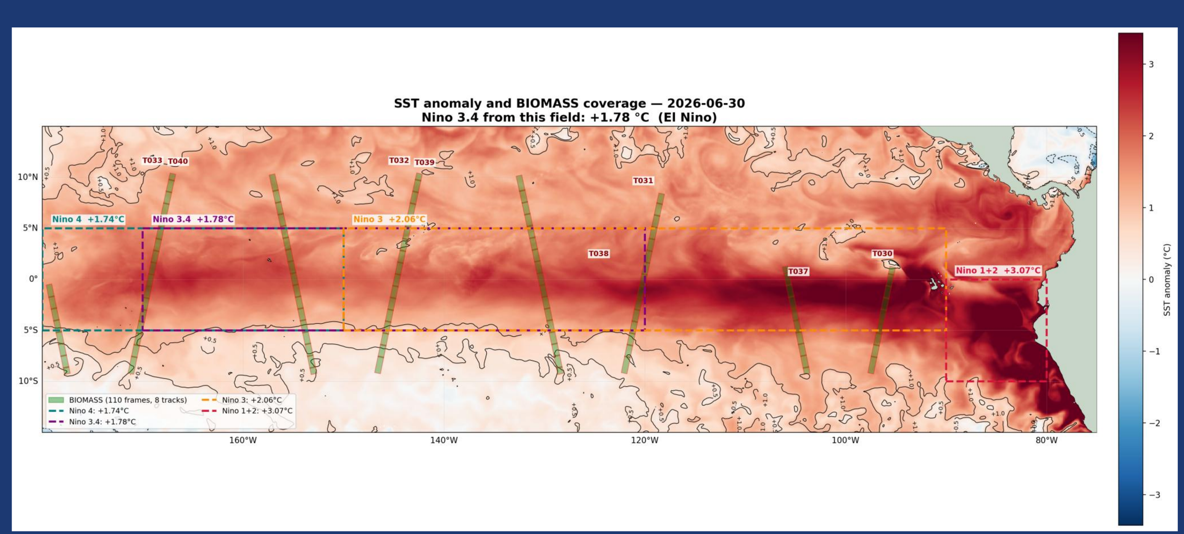


Fig 10. El-Niño study case, showing all swath availability overlaid on Sentinel-3 temperature plot. (Acquisition 30/06/26)

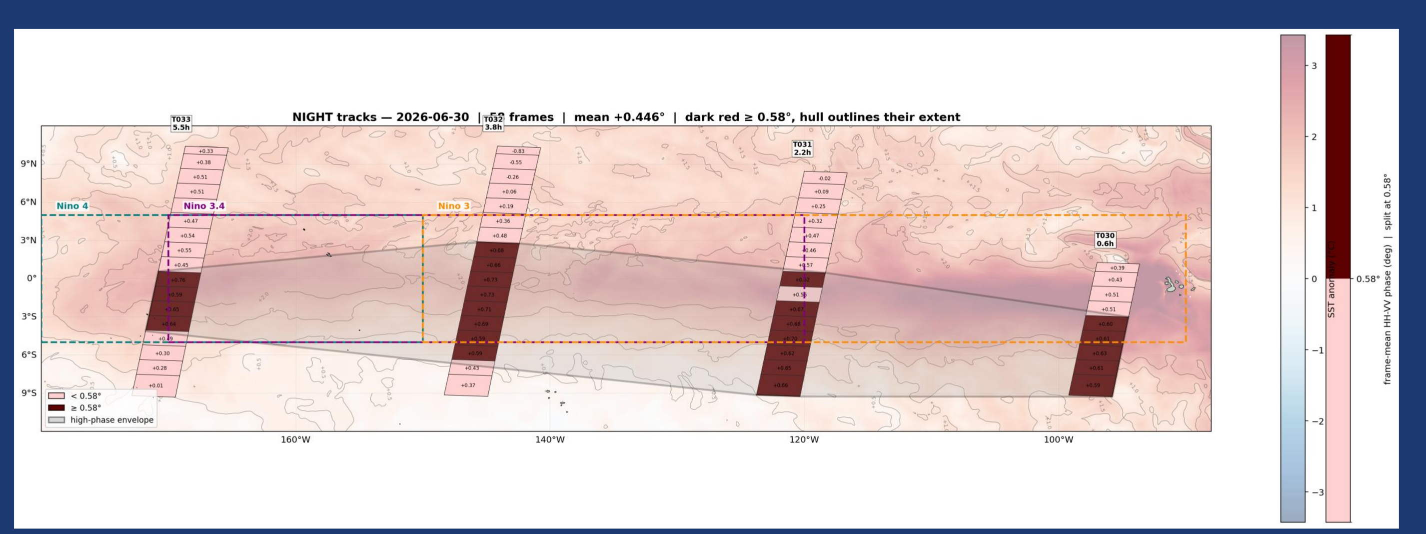


Fig 11. Product CPD mean mosaic displayed approximately mimicking El-Niño SSS & SST tongue.

References;
ESA Biomass Mission [1]
DLR Earth Observation [2]
Taillade T., Engdahl M., Fernandez D. (21 June 2023) IEEE Geoscience and Remote Sensing Letters (Vol: 20) [3]

Acknowledgements
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